

Task 59: Keys and Rooms

Problem Description

There are n rooms labeled from 0 to $n - 1$ and all the rooms are locked except for room 0. Your goal is to visit all the rooms. However, you cannot enter a locked room without having its key. When you visit a room, you may find a set of distinct keys in it. Each key has a number on it, denoting which room it unlocks, and you can take all of them with you to unlock the other rooms. Given an array `rooms` where `rooms[i]` is the set of keys that you can obtain if you visited room i , return `true` if you can visit all the rooms, or `false` otherwise.

Java Solution

```
class Solution {
    public boolean canVisitAllRooms(List<List<Integer>> rooms) {
        boolean[] visited = new boolean[rooms.size()];
        visited[0] = true;
        Stack<Integer> stack = new Stack<>();
        stack.push(0);
        int count = 1;

        while (!stack.isEmpty()) {
            int curr = stack.pop();
            for (int key : rooms.get(curr)) {
                if (!visited[key]) {
                    visited[key] = true;
                    stack.push(key);
                    count++;
                }
            }
        }

        return count == rooms.size();
    }
}
```